AXEL COLMANT

Research Engineer - Computer Vision & Embedded AI

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EXPERIENCE

AI & Software Engineer @ Verréo

Sep. 2025 – Present Céret, FRANCE

Full-stack development of the sorting machine's core software and design of a multimodal identification system.

- Architected a **multimodal recognition pipeline** combining OCR (Text) and **DinoV2** embeddings (Visual) to identify manufacturers.
- Implemented a **Vector Search** engine (Faiss/Qdrant) and a **Decision Fusion** logic to arbitrate between semantic parsing and visual similarity.
- Developed the machine's central control software, managing **SQL databases**, hardware synchronization (triggers), and real-time inference (YOLO-seg).

Multimodal AI | DinoV2 / Transformers | Vector Search (RAG) | OCR & NLP | Software Architecture

Computer Vision Research Engineer Intern @ Verréo

Apr. 2025 – Aug. 2025 Céret, FRANCE

Led a comparative study of deep learning architectures to solve fine-grained texture classification on cork surfaces.

- Conducted a systematic benchmark of dozens of model variants (ResNet, EfficientNet, DeepTen) combined with attention modules (CBAM).
- Analyzed the accuracy-latency trade-off to select the optimal architecture for edge deployment.
- Successfully transitioned the best-performing model from research to production, integrating it into the industrial sorting pipeline.

Model Benchmarking | Architecture Search | Attention Mechanisms | PyTorch | MLOps | Research to Production

Apprentice Software Engineer @ Le SAS Escape Game

Oct. 2021 – Aug. 2024 Béziers, FRANCE

Engineered a distributed IoT ecosystem coordinating networked microcontrollers and sensors.

- Architected a centralized management software in **Python** for real-time device orchestration.
- Developed embedded firmware (**C++/Arduino**) and implemented low-latency communication protocols (MQTT, Serial).

Python | Qt | IoT | Embedded Systems | Real-Time Control

Apprentice Electrical Engineer @ REEL International

Sep. 2019 – Jun. 2021 Saint-Cyr-au-Mont-d'Or

- Developed a **Python/Qt** software prototype to automate industrial costing workflows.
- Implemented communication gateways (**Ethway-Modbus**) for nuclear facility equipment.

gateways

Python | Qt | Modbus | Ethway | Industrial Automation

SKILLS

Python
C/C++

PyTorch
Computer Vision
Deep Learning
IoT/Embedded AI

MLOps
SQL
ROS
Gazebo
Linux / Bash
Git / CI
TensorRT

LANGUAGES

French
English

STRENGTHS

Autonomy | Team Spirit | Scientific Watch | Problem Solving | Intellectual Curiosity

EDUCATION

Master 2 Computer Science, specialization in Artificial Intelligence @ Université Claude Bernard Lyon 1

Sep. 2024 – Aug. 2025 Lyon, FRANCE

Master, Networks and Connected Objects, specialization in Robotics (CNAM) @ IMERIR

Sep. 2022 – Aug. 2024 Perpignan, FRANCE

Bachelor, Computer Science specialization in Robotics (CNAM) @ IMERIR

Sep. 2021 – Aug. 2022 Perpignan, FRANCE

DUT Electrical Engineering & Industrial Computing @ Université Grenoble Alpes

Sep. 2019 – Jun. 2021 Grenoble, FRANCE

PROJECTS

GPT-2 Implementation from Scratch @ Université Claude Bernard Lyon 1

2025

Implemented the full Transformer decoder architecture (OpenAI GPT-2) in raw PyTorch.

- Coded **Multi-Head Self-Attention**, Layer Normalization, and Positional Embeddings manually (no high-level wrappers).
- Trained the model on text corpora (Shakespeare) to demonstrate coherent character-level generation and context management.

LLM Architecture | Self-Attention | PyTorch | NLP

Swarm Intelligence Exploration @ Université Claude Bernard Lyon 1

2024

Developed a simulation of autonomous robots collecting resources using **Ant Colony Optimization** (Pheromones).

- Programmed decentralized logic where agents deposit digital pheromones to optimize retrieval paths.
- Solved the "coordination problem" without a central server, mimicking biological swarm behavior.

Multi-Agent Systems | Swarm Intelligence | Path Optimization | Emergent Behavior

Lunar Lander @ Université Claude Bernard Lyon 1

2024

Trained a reinforcement learning agent to land a spacecraft autonomously.

- Implemented Deep Q-Learning with custom reward shaping and network optimization.
- Analyzed convergence and generalization performance.

Reinforcement Learning | Deep Q-Learning | Gymnasium | Optimization

Autonomous Mobile Robot @ IMERIR

2023

Engineered a hybrid navigation system combining classical robotics with deep learning control.

- Orchestrated the **ROS Navigation Stack**, utilizing **SLAM** for mapping and **AMCL** for probabilistic localization.
- Integrated a **CNN model** (trained on MNIST) to drive navigation decisions based on real-time visual marker recognition.

ROS / Gazebo | SLAM / AMCL | Deep Learning (CNN) | Visual Servoing